Live Class



Topic : Pipes and Cistern

Zero to Hero Series

Pre-Assessment

TCS / Elitmus :

Question :

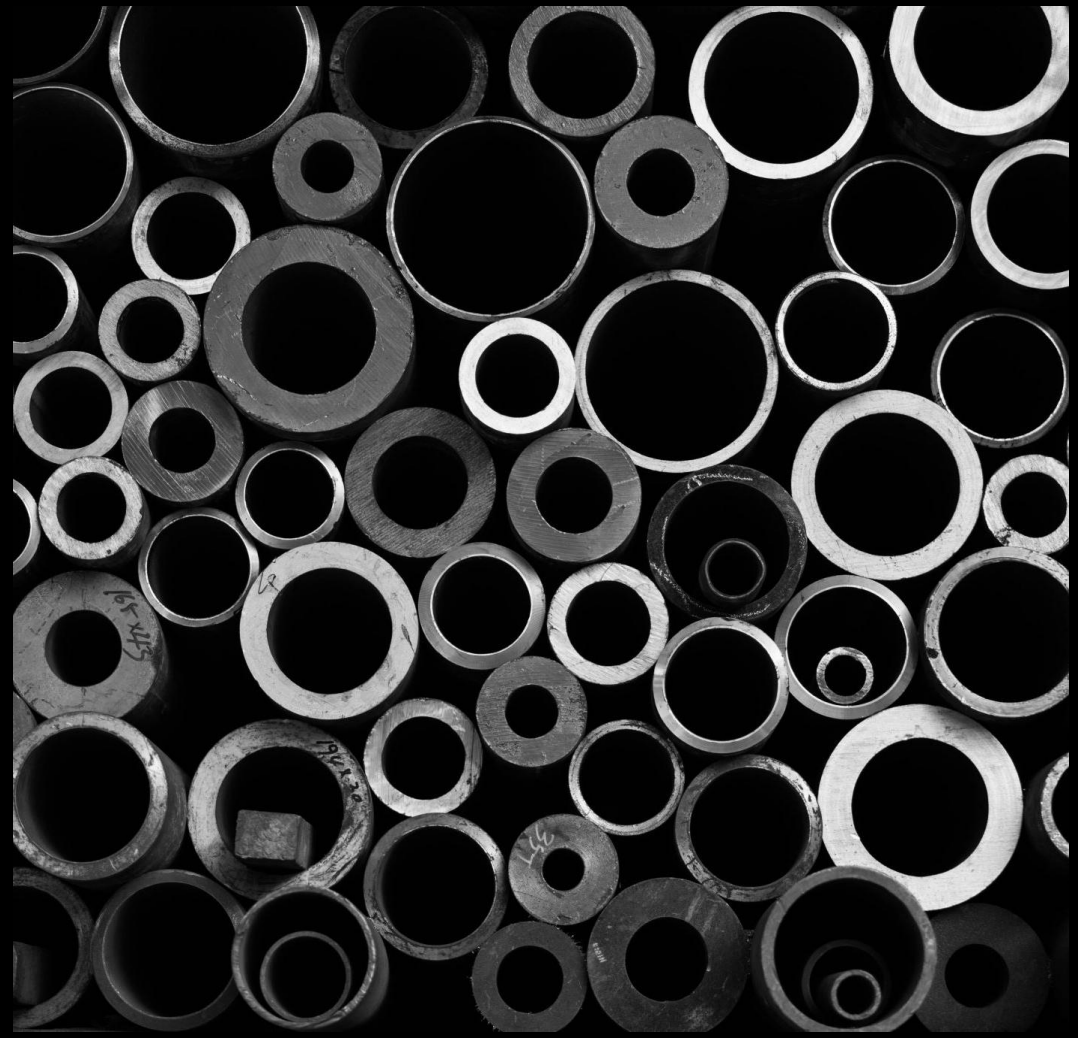
Pipe A can fill a tank in 12 hours. Pipe B can fill $33\frac{1}{3}$ % part of the same tank in 6 hours, whereas pipe C alone can empty full tank in x hours. When all the three pipes are opened together, $\frac{13}{15}$ part of the tank is filled in 12 hours. How much time (in hours) will A and C together take to fill 40% part of the tank?

TITA

Concept

- Which of these pipes will empty a pool the fastest?

1.One pipe of diameter 60 cm.
2. Two pipes of diameter 30 cm each.
3.Three pipes of diameter 20 cm each.



Revision:

Question 1:

A, B and C can complete a piece of work in 24, 6 and 12 days respectively. Working together, they will complete the same work in ?

Question 2:

A and B can do a work in 12 days, B and C in 15 days and C and A in 20 days. If A, B and C work together, they will complete the work in :

(1) 5 days (2) 10 days (2) 7 5/6 days (4) 15 2/3 days

Question 3:

A and B can do a work in 12 days, B and C in 8 days and C and A in 6 days. Then B alone will complete the work in how many days :

Question 4:

A can do 1/4 of a work in 3 days and B can do 1/6 of the same work in 4 days, how much will A get if both work together and are paid Rs. 180 in all?

1 : Rs 150

2 : Rs 100

3 : Rs 120

4 : Rs 160

5 : None of these

Question 5:

Sakshi can do a piece of work in 20 days. Tanya is 25% more efficient than Sakshi. The number of days taken by Tanya to do the same piece of work is:

a) 15

b) 16

c) 18

d) 25

e) None of these

TYPE 1: Foundation

Question 1 :

A cistern can be filled with water by a pipe in 5 hours and it can be emptied by a second pipe in 4 hours. If both the pipes are opened when the cistern is full, the time in which it will be emptied the cistern.

TITA

Question 2:

Pipe A can fill an empty tank in 18 hours and pipe B can fill the same empty tank in 24 hours. If both the pipes are opened simultaneously, how much time (in hours) will they take to fill the empty tank?

TITA

Question 3:

Pipe A can fill a tank in 12 minutes; pipe B can fill it in 18 minutes, while pipe C can empty the full tank in 36 minutes. If all the pipes are opened simultaneously, how much time will it take to fill the empty tank completely?

TITA

TYPE 2:

Question 4:

A pipe can fill a tank in 30 hours. Due to a leakage at the bottom, it is filled in 50 hours. How much time will the leakage take to empty the completely filled tank?

TITA

Question 5:

A pipe can fill a tank with water in 3 hours. Due to a leakage in bottom, it takes $3\frac{1}{2}$ hours to fill it. In what time the leak will empty the fully filled tank?

TITA

Question 6 :

A tap can fill a cistern in 8 h and another tap can empty it in 16 h. If both the taps are open, the time (in hours) taken to fill the tank will be

TITA

Question 7:

Two pipes can fill a tank in 10 hours and 16 hours respectively. A third pipe can empty the tank in 32 hours. If all the three pipes are opened simultaneously then in how much time the tank will be full? (in hours)

TITA

TYPE: 3

Question 8:

Pipes A and B can fill a tank in 16 hours and 24 hours, respectively, whereas pipe C alone can empty or the full tank in x hours. When all the 3 pipes are opened together, the tank is full in $20\frac{4}{7}$ hours. What is the value of x?

TITA

Question 9:

Pipe A can fill a tank in 12 hours. Pipe B can fill $33\frac{1}{3}\%$ part of the same tank in 6 hours, whereas pipe C alone can empty full tank in x hours. When all the three pipes are opened together, $\frac{13}{15}$ part of the tank is filled in 12 hours. How much time (in hours) will A and C together take to fill 40% part of the tank?

TYPE: 4

Question 10:

Two pipes can fill a tank with water in 15 and 12 hours respectively and a third pipe can empty it in 4 hours. If the pipes be opened in order at 8, 9 and 11 a.m. respectively, the tank will be emptied at

(a) 11 : 40 a.m. (b) 12 : 40 p.m.

(c) 1 : 40 p.m. (d) 2 : 40 p.m.

(e) none of these

Question 11:

Three pipes A, B and C can fill a tank in 6 hours, 9 hours and 12 hours respectively. B and C are opened for half an hour, then A is also opened. The time taken by the three pipes together to fill the remaining part of the tank is:
(a) 3 hours (b) 2 hours (c) 2.5 hours (d) 3.5 hours (e) none of these

TYPE: 5

Question 12:

Each inlet pipe can fill an empty cistern in 84 hours while each drain pipe can empty the same cistern from a filled condition in 105 hours. When the cistern is empty, 9 inlet pipes and 10 outlet pipes are simultaneously opened. After how many hours will the cistern be completely filled?

TITA

Question 13:

An inlet pipe can fill an empty tank in 120 hours while an outlet pipe drains a completely-filled tank in 54 hours.
If 8 inlet pipes and 3 outlet pipes are opened simultaneously, when the tank is empty, then in how many hours will the tank get completely filled?

(a) 81
(b) 96
(c) 72
(d) 90

TYPE: 6

Question 15:

Two pipes X and Y can fill a cistern in 24 minutes and 32 minutes respectively. If both the pipes are opened together, then after how much time (in minutes) should **Y be closed so that the tank is full in 18 minutes?**